

VR. A Formal System — v1.1.2 (revision note)

An added clarification of the ontological status of the nullary base (operational reading)

Vitaly Reznik

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Scope. This is the revision delta for *VR. A Formal System* v1.1.2. *No construction, definition, theorem, or axiom-profile is altered.* The single change is an added clarification — an interpretive-layer paragraph — stating the ontological status of the nullary base *positively*. v1.1.1 already says, correctly, that “ $\emptyset$ is a nullary operation, not a posited thing; there are only operations”. That is the negative half (no ontology of substance); this note adds the positive half, so the reading is not mistaken for a claim of *zero* ontology.

The clarification (operational reading — the status of the base)

Stating only that $\emptyset$ is “not a posited thing” leaves open the natural objection: to posit that there *is* an operation is itself a commitment. The corrected reading owns this. Add, as an operational-reading remark:

The claim “there are only operations; $\emptyset$ is a nullary operation, not a posited thing” removes any ontology *of substance* — no object that *is* prior to operations. It does not claim the absence of ontology altogether. To posit that there is an operation — that there is doing — is itself a minimal ontological commitment, and VR makes exactly one: this single act. The commitment is not arbitrary but *self-affirming*: to deny that there is operation is already to perform one (the denial). VR’s base is therefore best described not as “no ontology” but as the *thinnest, self-affirming* ontology — one act, “there is doing”, nothing further presupposed.

A fuller transcendental treatment (the retortion structure; and why grounding the operation in a *physical* fact does not help — it only trades a thin commitment for the heavier one of physicalism, and makes necessary arithmetic hostage to contingent physics) is deferred to the accompanying philosophical essay, not carried in this formal work.

Why VR. A Formal System is otherwise unaffected

The change is interpretive only. The formal content — the free term-algebra on $\{\emptyset$ (nullary), t (unary) $\}$ with the recursor, membership defined, A3 (succession) and A4 (induction) as theorems, and the axiom-free profile $[]$ — is untouched, as is the Lean module `VRCycle.VR`. The nullary base $\emptyset$ is characterised exactly as before by $\forall x \neg(x \in \emptyset)$; only its *reading* is made precise. This note refers to *VR. A Formal System* v1.1.1 (Zenodo DOI 10.5281/zenodo.20688164; concept 10.5281/zenodo.20212091). *Prepared with the assistance of Claude (Anthropic, Opus 4.8) in the cycle’s Variant A architecture; all mathematical content and positions are due to the author, Vitaly Reznik.*